Sahil Bharti, PhD

Assistant Professor— Mechanical & Aerospace department
School of Engineering, Chanakya University, Bengaluru, India
PhD Alumnus, Indian Institute of Technology Madras
sahibhart@gmail.com
ResearchGate

Professional Summary

Current Assistant Professor at Chanakya University and PhD in Mechanical Engineering with expertise in robot-assisted sheet metal forming, highly relevant to aerospace and automotive applications. Published researcher with 6+ peer-reviewed papers in high-impact journals, including Nature Scientific Reports, and 3 patents. Seeking to leverage my research expertise to drive innovation in industrial automation.

APPOINTMENTS

Assistant Professor Feb 2026 - Present
Chanakya University, Bengaluru

Delivering core B.Tech courses, including Engineering Mathematics (Probability and Statistics) and Engineering Mechanics. Responsible for complete course planning, curriculum development, and creating comprehensive student evaluation materials.

EDUCATION

PhD Mechanical Engineering (CGPA 8.75/10) 2017-2024
Indian Institute of Technology Madras

"Fundamental studies on toolpath and Heat-assisted methods for formability analysis and development of the Roboforming process."

(Thesis defended on 20 Dec 2024)
Advisors: Dr. K Hariharan (IITM) / Dr. S K Panigrahi (IITM)
Key Achievements:

1. Published 6 peer-reviewed papers, including Nature Scientific Reports
2. Dissertation resulted in 3 patents and 137.5 L Industry funding

M.Tech Mechanical Engineering (CGPA 8.46/10) 2015-2017
Indian Institute of Technology Gandhinagar

BE Production Technology (CGPA 8.01/10) 2009-2013
Madras Institute of Technology, Anna University

RESEARCH AREAS

Plasticity and sheet metal forming

- Phenomenological damage model
  - Evaluated the applicability of the **phenomenological damage models** for formability assessment in incremental sheet metal forming (ISF); the importance of path effect and friction was highlighted.
  - Proposed an analytical methodology to evaluate damage progression in ISF outside the FE environment **(ABAQUS/Explicit); effectively achieved reduced computational time up to half**.
  - Proposed dislocation density as a damage variable for uncoupled models. Strain rate and temperature effect under proportional loading can be accommodated in the approach.
- Fracture Forming Limit Diagram (FLD)
  - **Fracture forming limits (FFL)** in ISF are developed experimentally and numerically. The obtained forming limits are higher than the conventional necking-based forming limit curves.

Robot assisted sheet metal forming

- Highlighted the **influence of component profile asymmetry and tool path method in formability** in incremental sheet metal forming.
- Demonstrated the **effect of local friction heat for hybrid roboforming** process.
- Proposed a hybrid methodology including **experimental, numerical, and analytical methods** to highlight the possible error-causing factors in **geometric accuracy in robot-assisted incremental forming**.

Finite Element modelling

- Simulate **fracture behavior** in sheet metal forming using Finite Element analysis using the continuum damage approach.
- Analyse the **thermo-mechanical behavior of Magnesium alloy** at a higher temperature using Finite Element Method (FEM) model - Achieved higher formability and proposed a comprehensive understanding of the underlying mechanisms behind the observed improvements.
- Proposed methodology to estimate the **contribution of tool deviation, springback, and machine compliance on the geometric error** in ISF.

Bulk metal Die-less forming

- Designed and developed an experimental setup to study and demonstrate **die-less bulk metal forming** of mild steel rod, assisted with localized induction-based intermittent heating and cooling.

Mechatronics

- Designed and developed a **universal cleaning machine prototype** to automate the conventional surface cleaning of the horizontal and vertical surfaces aimed for high-rise buildings - mimicking the wetting, sweeping, and drying stages.

PATENT

1. Hariharan Krishnasamy, Anuj k Tiwari, Sahil Bharti and Srivardhan Reddy Palwai "Method for development of Preform shapes for incremental forming and Robo-forming applications and preforming devices". Application No. 202441025143
2. Eldho Paul, Sahil Bharti and Hariharan Krishnasamy, "A System for Robot-Assisted Incremental Manufacturing Process, and Methods Thereof ". Application No. 202441031550
3. Eldho Paul, Sahil Bharti and Hariharan Krishnasamy, "Self-Lubricating Forming tool for Roboforming." Application No. 202441066634

PEER-REVIEWED JOURNAL PAPERS [Google Scholar]

6. Srivardhan Reddy Palwai, Sahil Bharti, Anuj Tiwari, Hariharan Krishnaswamy, Saravana Kumar Gurunathan 'Mathematical approach to design preform for multi-stage robot-assisted incremental forming'International Journal of Material Forming,18,Article number: 62,,July 2025
5. Sahil Bharti, Eldho Paul, Anandu Uthaman, Alexandr Klimchik, Hariharan Krishnaswamy, Riby Abraham Boby 'Systematic analysis of geometric inaccuracy and its contributing factors in roboforming'Nature Scientific Reports 14Article number: 20291 2024
4. Sahil Bharti, Karthik Subramanya Karvaje, Hariharan Krishnaswamy, Anupam Agrawal and Sushnta Kumar Panigrahi 'Investigation of feature-based and space-filling tool path strategies for formability in incremental sheet metal forming'The International Journal of Material Forming 16,60 2023
3. Narinder Kumar, Sahil Bharti Hariharan Krishnaswamy and Anupam Agrawal 'Exploring deformation mechanics of temperature assisted incremental forming with hybrid heating'Journal of Manufacturing Processes 104,472-484 2023
2. Hariharan Krishnaswamy, Vikram Balaji and Sahil Bharti 'A new path independent variable to model ductile damage'Materials Science and Technology 39,496-500 2023
1. Sahil Bharti, Aishwary Gupta, Hariharan Krishnaswamy, Sushanta Kumar Panigrahi and Myoung Gyu Lee 'Evaluation of uncoupled ductile damage models for fracture prediction in incremental sheet metal forming' CIRP Journal of Manufacturing Science and Technology 37,499-517 2022

CONFERENCE PAPERS

4. Srivardhan Reddy Palwai, Sahil Bharti, Anuj Tiwari and Hariharan Krishnaswamy, "

   A Novel Method to Design Preform Shape for Robo-forming", *Conference: 2024 10th International Conference on Control, Decision and Information Technologies (CoDIT)* **July 2024**, IEEE (DOI: 10.1109/CoDIT62066.2024.10708559).
3. Eldho Paul, Sahil Bharti, Anandu Uthama, Riby Abraham Boby, Hariharan Krishnaswamy, and Alexandr Klimchik 'Evaluation Of Deviations Due To Robot Configuration For Robot-based Incremental Sheet Metal Forming' Advances in Robotics 2023, 6TH International conference of the Robotics Society JULY 5–8 IIT Ropar **July 2023**

2. Sahil Bharti, Deepu Raneesh, Hariharan Krishnaswamy, Sushanta Kumar Panigrahi , " Analytical approach to damage prediction in incremental sheet metal forming", *International Deep-Drawing Research Group Conference (IDDRG 2022)* **June 06-10**, Lorient, France (2022). (In IOP Conference Series: Materials Science and Engineering, Volume 1238, 012024)

1. Eldho Paul, Hariharan Krishnaswamy, Riby Abraham Boby and Sahil Bharti "A Finite Element Analysis Study on the Effect of Tool Stiffness in Incremental Forming Process" The 6th Joint International Conference on Multibody System Dynamics and The 10th Asian Conference on Multibody System DynamicsOctober 16-20New Delhi, India **Nov 2020**

LAB DEVELOPMENT ACTIVITIES

@ IIT Madras:

a) Department of Heavy Industry and Mahindra & Mahindra (DCAAI Grant), " 3D forming of sheet metal for industrial applications", Rs.137.5 lakhs (PI, 2019-2021).**Completed**

i. Designed and developed a **Robot-assisted incremental forming** method and successfully set up an experimental facility at IIT Madras.

ii. Developed a **vertical sheet metal fixture** for full-scale commercial, automotive sheet metal parts using Robot-assisted forming.

iii. Bench-marked roboforming process for automotive components

AWARDS AND RECOGNITION

- Received a **pre-incubation startup grant** at IIT Madras.
- Received **Young Achievers Award 2017** from Madras Institute of Technology Alumni Association for excellence in research performance, covered by national Daily, TIMES of INDIA
- Recipient of **International Emersion Experience Award** by IIT Madras as a travel grant worth 3 Lakhs INR.
- Published article in the international journal of "South Asian Research Center" and received the **excellent paper award** at the " International Conference on Computer Science and Mechanical Engineering."

SKILLS

- **Programming:**MATLAB, Python, C/C++
- **Software:** ABAQUS, RoboStudio, Fusion 360, LATEX, Solidworks, MASTERCAM, POLYWORKS Inspector, Microsoft Office suite, Arduino, ANSYS
- **Experimental techniques:**
  - **Formability analysis** -Forming limit curves, Fracture forming limits, grid circle analysis
  - **Metal forming** - Deep drawing, Extrusion, punching

  - **Machining**- CNC Milling, Drilling
  - **Mechanical characterization**- Tensile testing, bulge testing, 3D scanning, Digital image correlation
- **Languages known:** English, Hindi, German (A1)

TEACHING EXPERIENCE

Assistant Professor: Chanakya University, Bengaluru: Feb 2026 - Present

- **Core courses- B.Tech**
  a) **Engineering Mathematics: Probability and Statistics**, Feb-May 2026
  b) **Engineering Mechanics**, Feb-May 2026
  c) **Robotics and Automation Systems**, July 2026- ongoing
  d) **Physics for Engineers**, July 2026- ongoing

Teaching Assistant: Indian Institute of Technology Madras: 2017-2025

- **Core courses- B.Tech**
  a) **Metal forming analysis and tool design** (ME5012), Jan-May 2019
  b) **Engineering Drawing** (ME1120), Jul-Dec 2021
  c) **Study and Programming of Industrial Robot** (ME3484), Jan-May 2020, 2022
  d) **Measurements,Instrumentation and Control** (ME2400), Jan-May 2021

Teaching Assistant: Indian Institute of Technology Gandhinagar: 2015-2017

- **Core courses- B.Tech**
  a) **Manufacturing Processes**(ME331), Jan-May 2015, 2016